PROJECT TITLE: SCAMSHIELD MY

Explainable, Community-Verified Scam Intelligence and Intervention Platform

1. Problem Statement

Digital scammers impersonate banks, government agencies, employers, delivery services, marketplaces and family members. They use urgency, fear, secrecy, authority and attractive rewards to pressure users into clicking links, scanning QR codes, sharing credentials or transferring money before verifying the request.

Existing protection is fragmented. Users may need to search a phone number, inspect a domain, read official advisories, ask other people and report the incident through different channels. A simple scam/not-scam label is also insufficient because false positives create unnecessary fear while false negatives may create false confidence.

Key gaps

- Suspicious text, screenshots, URLs, QR codes, phone numbers and payment details are often checked separately.
- Most tools provide a label or score without clearly explaining the manipulation or technical warning signs.
- New campus or community scam patterns may spread before static databases are updated.
- Users may not know the safest next action or how to report evidence without exposing personal information.
- Older adults and users with lower digital confidence need larger text, plain language and optional guardian support.

> **Proposed opportunity**
> Create one explainable workflow that analyses suspicious content, highlights the evidence, recommends safer actions and uses moderated reports to build local scam intelligence.

2. Objectives, Users and Innovation

Project objectives

1. Develop a focused prototype that analyses text, screenshots, URLs and QR codes within a two-week sprint.
2. Generate explainable risk results that identify the main suspicious indicators in plain language.
3. Recommend safe verification, blocking, evidence-preservation and reporting actions without guaranteeing a final judgement.
4. Demonstrate a basic moderated reporting workflow that converts reviewed cases into local scam intelligence.
5. Measure basic technical performance, usability and decision improvement using a small controlled test group.

Target users and needs

User group	Main need	Relevant capability
University students	Fast checks for job, parcel, scholarship, rental and marketplace scams	Mobile quick check, URL/QR analysis and reporting
Older adults	Simple guidance for impersonation, investment and family-emergency scams	Large-text mode, step-by-step actions and optional guardian support
Family caregivers	Help a relative review a high-risk request without continuous monitoring	Consent-based sharing and action checklist
Campus units	Identify recurring local scams and issue controlled alerts	Moderator queue, trend dashboard and alert approval
Trusted reviewers	Confirm, reject, merge or escalate reports	Evidence review, reputation context and audit log

Innovation and differentiation

- Explainable prevention: shows exactly which social-engineering and technical indicators increased risk.
- Multi-format evidence fusion: combines text, OCR, URL/QR features, payment context, reputation and community evidence.
- Community-verified intelligence: reviewed reports can strengthen confidence and contribute to local trends without publishing unverified accusations.
- Action-oriented intervention: every result includes a verification pathway and practical next steps.
- Inclusive design: large text, plain language, clear navigation and optional guardian support.
- Campus-to-national scalability: the same core workflow can later support multiple institutions and partners.

3. Proposed Solution, Scope and Workflow

Core solution

Capability	What the system does
Analyse	Accepts pasted text, screenshots, URLs, QR codes, phone numbers and payment-request details.
Explain	Highlights urgency, impersonation, credential requests, suspicious domains, payment anomalies and inconsistencies.
Act	Recommends official verification, pausing payment, blocking, preserving evidence, guardian support or reporting.
Verify	Allows authorised reviewers to confirm, reject, merge, escalate or request more evidence.
Learn	Uses confirmed cases to update anonymised trends, educational cards and the local knowledge base.

Two-week prototype scope

- Core user flow: text paste, screenshot upload, QR image upload, URL input and an explainable result page.
- Core analysis: OCR, QR decoding, URL parsing, transparent scam rules and a lightweight classifier if time permits.
- Explainable result: risk band, confidence, highlighted evidence, uncertainty statement and recommended actions.
- Basic reporting demo: redacted report submission, simple case status and moderator confirmation or rejection.
- Simple dashboard: a small moderator queue and prepared anonymised trend summaries; advanced analytics are stretch work.
- Accessibility: plain-language results, larger text and clear buttons; read-aloud and guardian alerts are stretch features.

End-to-end workflow

1 Submit	2 Extract	3 Analyse	4 Explain	5 Act	6 Verify & learn

Out of scope for the MVP

- Automatic access to private WhatsApp, Telegram or encrypted messages.
- Guaranteed scam detection, automatic transaction blocking or autonomous legal/financial advice.
- Deepfake voice/video detection and nationwide law-enforcement case management.
- Unmoderated public accusations or public display of identifiable personal evidence.
- Full production integration with banks, telcos or government systems.

4. Technical and Responsible Design

Technical approach

Layer	Recommended approach
User interfaces	Flutter mobile app; responsive web checker and moderator dashboard
Backend	FastAPI or Node.js REST services with authentication, consent controls and rate limits
Data	PostgreSQL for cases and audit records; private object storage for screenshots
Input processing	OCR for screenshots, QR decoding and URL/phone/payment-pattern extraction
Risk engine	Transparent rules first, optional lightweight classifier, URL/QR features and prepared local reputation data
Deployment	Docker on a student-friendly cloud platform; local fallback data for demonstrations
Analytics	Simple dashboard for confirmed categories, channels, duplicate cases and time-based trends

Risk scoring and explainability

The final risk result should not be decided by a generative model alone. The system combines transparent rules, a lightweight text classifier, URL/QR indicators, contextual inconsistencies, reputation and verified community evidence. Each result displays a separate risk band and evidence confidence.

Risk band	User-facing guidance
Low evidence	No strong indicator detected; verify independently before acting.
Caution	Suspicious elements found; pause and use an official verification channel.
High risk	Multiple scam indicators; do not pay or share credentials until independently verified.
Critical	Strong combined evidence or verified local matches; block, preserve evidence and report.

Privacy, cybersecurity and ethics

- Collect only the content required for the selected check; allow analysis without forcing a public report.
- Show a redaction preview and mask names, phone numbers, account details and unrelated chat content.
- Use encrypted transport, role-based access, strong moderator authentication, private storage and audit logs.
- Never open suspicious URLs automatically; analyse structure or metadata using a safe process.
- Require human review before public alerts or accusations; keep raw evidence restricted.
- State clearly that results are preliminary guidance and do not replace official verification.

5. Validation, Impact and UCRIX Alignment

Validation plan

- Problem validation: conduct a small number of short interviews or feedback sessions with accessible target users.
- Technical testing: measure OCR, QR decoding, URL extraction, classification performance, latency and failures.
- Usability testing: observe first-time users completing the scan, explanation and reporting flow without coaching.
- Decision impact: compare intended action before and after users view the explanation and safe-action guidance.
- Community workflow: test the basic report-review process and ensure unverified reports are not published.
- Accessibility and security: check comprehension, text size, contrast, file validation, redaction and restricted moderator access.

Suggested prototype KPIs

Impact area	Target
High-risk identification	At least 85% recall on a controlled labelled test set
False-warning control	At least 80% specificity on legitimate test cases
Explanation comprehension	At least 80% of users identify the key warning signs
Decision improvement	Target 30 percentage-point improvement towards safer intended actions
Usability	At least 85% complete the primary scan without assistance
Speed	Median result below five seconds for common prototype cases
Privacy	At least 95% successful redaction in prepared cases
Awareness	At least 20% average improvement in a short pre/post assessment

UCRIX evaluation alignment

Criterion	Evidence to present
Relevance	User interviews, local scenarios and a clear before-loss problem story
Innovation	Explainability, evidence fusion, community verification and action guidance
Technical execution	Working OCR/QR/URL analysis, risk engine, dashboard and access controls
Measurable impact	Confusion matrix, pre/post decisions, task completion and KPI dashboard
Feasibility	Strict two-week MVP, RM1,000 maximum planning budget and fallback plans
Demo and scalability	Three-minute end-to-end demonstration and phased university-to-partner roadmap

Note: all targets are internal prototype goals and should be adjusted after baseline testing. Results must be reported honestly with sample size and limitations.

6. Feasibility, Team, Two-Week Timeline and Budget

Student-team feasibility and roles

Role	Main responsibility
Project lead / product manager	Scope, stakeholder interviews, requirements, sprint coordination and pitch integration
AI / data engineer	Dataset, rules, classifier, risk fusion, explanations and model evaluation
Backend / security engineer	APIs, database, authentication, storage, moderation, privacy and logging
Frontend / UX engineer	Mobile/web interface, elderly mode, user journeys and accessibility testing
Full-stack / research lead	Dashboard integration, analytics, user testing, documentation and demo preparation

Two-week development roadmap

Period	Focus and output
Days 1-2	Freeze the core scope, confirm user flow, prepare sample scam/legitimate cases, wireframes, task assignments and repository.
Days 3-5	Build the user interface, backend endpoint, text/URL input, screenshot OCR and QR decoding.
Days 6-7	Implement transparent risk rules, result bands, evidence highlighting and safe-action guidance; integrate the core flow.
Days 8-10	Add basic report submission, moderator review, simple trend display and accessibility improvements where feasible.
Days 11-12	Run technical and usability tests, fix critical issues, refine explanations and prepare fallback data.
Days 13-14	Finalise the proposal, poster/pitch, recorded backup, live demonstration and team rehearsal.

Estimated prototype budget

Item	Estimated RM
Cloud hosting and database	200
Optional OCR/API testing credits	150
Domain and basic deployment	100
User testing materials and transport	150
Pitch and demonstration setup	150
Contingency	250
Estimated total	1,000

Feasibility boundary
The two-week prototype is achievable only with strict scope control. Core analysis and explainable results must be completed first; advanced community analytics, multilingual expansion, guardian alerts and production integrations remain stretch or future work. Free-tier and university resources may reduce the estimated cost.

7. Risks, Scalability, SDGs and Conclusion

Key risks and mitigation

Risk	Mitigation
False positives / negatives	Use evidence bands, balanced test cases, uncertainty wording and independent verification guidance.
Insufficient or biased data	Use curated, consented and redacted examples; version datasets and test across languages and user groups.
Fake or abusive reports	Apply reputation, rate limits, duplicate checks, moderator approval and appeals.
Privacy exposure	Redact sensitive fields, minimise retention, restrict evidence and log moderator access.
Two-week scope overrun	Freeze stretch features early, assign parallel tasks and protect the core end-to-end demonstration.
Demo or service failure	Use local fallback data, manual input fallback, a recorded backup and a rehearsed reset procedure.

Scalability and adoption pathway

Phase	Scope
1. University pilot	Student checker, campus moderators and a local trend dashboard
2. Multi-campus service	Shared knowledge base, institution-specific alerts and central quality control
3. Partner integrations	Banks, telcos and marketplaces contribute verified indicators and receive aggregated intelligence
4. Public platform	Broader multilingual access, guardian networks, official reporting links and advanced modules

Expected impact and SDG contribution

- Reduce unsafe clicking, credential sharing and payment decisions by helping users pause and verify.
- Improve digital literacy because every result explains the warning signs instead of showing only a label.
- Provide universities with anonymised local scam trends and a controlled community alert workflow.
- Support SDG 16 (trust and responsible reporting), SDG 9 (AI and cybersecurity innovation), SDG 10 (inclusive protection) and SDG 4 (digital-safety education).

Conclusion

ScamShield MY is strongest as an explainable intervention and community-intelligence platform, not as a perfect detector. Its competition value comes from a clear end-to-end story: suspicious content is analysed, the evidence is explained, the user chooses a safer action, a moderated report strengthens local intelligence and the community learns from confirmed patterns.

Recommended closing line
ScamShield MY does not ask users to trust a black-box label. It shows the evidence, slows down manipulation and connects one person’s warning to a safer community.

Source basis

Condensed from the complete ScamShield MY proposal and its source concepts: the UCRIX ScamShield AI proposal pack, the ScamRadar brainstorming summary and the UCRIX/URIIS project-selection framework. Architecture, targets, timeline and budget remain planning assumptions that require validation and testing.